The coordinate ring of the k-fold iterated commutator locus for 2×2 matrices

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Abstract

For 2×2 matrices $A_1, ..., A_k$, write $[A_1, ..., A_k]$ for the left-normed iterated commutator $[...[[A_1, A_2], A_3], ..., A_k]$, and I_k for the ideal, in the $3k$-variable reduced-coordinate polynomial ring R_k, cutting out its vanishing locus. We prove, for every $k \geq 2$ over any field of characteristic $\neq 2$, and as four independently-established results rather than one bundled claim: I_k has codimension 2; I_k has exactly 3 minimal generators; R_k/I_k is Cohen-Macaulay; and I_k is radical. The last of these, together with an explicit component count resting on a non-containment argument, assembles into the **Primary Decomposition Theorem**: $I_k = P_2 \cap \cdots \cap P_k$ is an irredundant primary decomposition into exactly $k-1$ primes, following an explicit recursive block-involvement pattern. The proof identifies I_k as the ideal of 2×2 minors of an explicit 2×3 matrix (a determinantal ideal, not merely determinantal-**looking**), and invokes classical determinantal-ideal theory (Bruns-Vetter) and an explicit rank-2 Jacobian witness on every component (Serre's criterion) for the algebraic and radicality halves respectively.

Keywords. Iterated commutator; determinantal ideal; commuting variety; Cohen-Macaulay ring; primary decomposition; radical ideal; cross product; $\mathfrak{sl}_2$.

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1. Introduction

2×2 matrices A, B have commutator $[A, B] = AB - BA$, and the **iterated** left-normed commutator of k matrices is $[A_1, ..., A_k] := [...[[A_1, A_2], A_3], ..., A_k]$, matching the standard lower-central-series convention for Lie algebras. The object arose for the author twice: in the fiber-counting theory of word maps $\pi : F_n \to G$, F_n the free group on n letters, after a close reading of Tambour [13] – and in the algebraic-complexity question of how many scalar multiplications are needed to **compute** $[A_1, ..., A_k]$, which remains open for $k \geq 3$. The one fact needed from that story is purely algebraic: an explicit bilinear formula for the ordinary 2×2 commutator, chained across k matrices, gives an explicit polynomial parametrization of $[A_1, ..., A_k]$'s reduced coordinates (Definition 2 below), and the subject of this article is the **vanishing locus** of that parametrization, as an algebraic variety, together with its coordinate ring. At $k = 2$ this locus **is** the classical commuting variety; for $k \geq 3$ the vanishing of the iterated bracket is weaker than pairwise commutation, so the locus studied here properly contains the pairwise-commuting one.

The classical object – $C_n := \{(A, B) \in M_n \times M_n : AB = BA\}$ for $n \times n$ matrices – has been studied since the 1950s. Its irreducibility was proven independently by Motzkin and Taussky [3] and by Gerstenhaber [5], the latter also establishing the bound $\dim_{\mathbb{F}} \mathbb{F}[A, B] \leq n$ for the subalgebra generated by two commuting matrices. Whether the coordinate ring $\mathbb{F}[A, B]/I$, I generated by the entries of $AB - BA$, is **reduced** and **Cohen-Macaulay** for every n is a much harder question, attributed to Michael Artin and Melvin Hochster [6] and stated precisely as "Conjecture 1.1" in Majidi-Zolbanin and Snapp [7], while Chau [8] records a dated (1984) reducedness conjecture of Hochster about the same loci. Hreinsdóttir [9] attacks the conjecture case by case at $n = 3, 4$; Knutson [10] studies the minimal primes of closely related schemes. At the time of writing, Cohen-Macaulayness is settled for $n \leq 4$ (by hand at $n \leq 2$, by machine computation at $n = 3, 4$ [7] [9]) and open for every $n \geq 5$; reducedness is settled for $n \leq 3$ and open for $n \geq 4$ [8].

At $n = 2$ – this article's $k = 2$ base case – the question is comparatively easy, and is answered again, as a special case, by Theorem 28 below: I_2 (Definition 2) is literally the ideal of the $n = 2$ commuting variety, expressed in reduced (trace-removed) coordinates. The genuinely new object is the **iterated** generalization to $k \geq 3$: not the classical commuting variety, but a structurally analogous determinantal ideal, built by chaining the same bilinear cross-product formula that governs the $n = 2$ case – and for it, the same two questions the Artin-Hochster conjecture asks, Cohen-Macaulayness and reducedness, are answered completely and unconditionally for every $k \geq 2$ (Theorem 41). In that sense the results below are a fully worked structural analogue of the still-open classical question, in a setting simple enough ($n = 2$, but k unbounded) to admit a complete answer.

2. Setup: the signed encoding and the defining ideal

Throughout, $\mathbb{F}$ is a field of characteristic $\neq 2$, and all matrices, vectors and polynomial rings are over $\mathbb{F}$ unless said otherwise; the displayed examples below are computed over $\mathbb{Q}$, but nothing in the definitions is special to it. One typographic convention is used throughout: an **overline** denotes a **vector** – an element of $\mathbb{F}^3$ (or, for R_k itself, of the free module R_k^3) – while the same letter, plain, with a two-index comma subscript, denotes one of that vector's individual scalar **components**. So $\overline{v}_i \in \mathbb{F}^3$ is block i's whole reduced-coordinate vector, with components $v_{i,1}, v_{i,2}, v_{i,3} \in \mathbb{F}$; likewise $\overline{w}_j$ versus $w_{j,1}, w_{j,2}, w_{j,3}$.

Step 1: discard the scalar part. Write $A = \left(\begin{smallmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{smallmatrix}\right)$ as the sum of its scalar part $(\mathrm{tr}(A)/2)I$ and its traceless remainder

$$A_0 := A - (\mathrm{tr}(A)/2)I = \begin{pmatrix} d_a/2 & a_{12} \\ a_{21} & -d_a/2 \end{pmatrix}, \quad d_a := a_{11} - a_{22}. \tag{1}$$

Scalar matrices are central, so they never contribute to a commutator:

$$[cI, B] = cIB - BcI = cB - cB = 0 \tag{2}$$

for every B and every scalar c, hence $[A, B] = [A_0, B_0]$ identically – no information relevant to **any** bracket is lost by discarding the scalar part, so from here on only the traceless remainder matters.

Step 2: encode the traceless remainder. A traceless matrix $\left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$ has three free entries. Rather than reading them off directly, take

$$\mathrm{enc}(A) := (a_{11} - a_{22}, a_{12} + a_{21}, a_{12} - a_{21}) \in \mathbb{F}^3, \tag{3}$$

the traceless $(1,1)$-entry doubled. On a traceless matrix written as above this reads

$$\mathrm{enc}\begin{pmatrix} x & y \\ z & -x \end{pmatrix} = (2x, y + z, y - z), \tag{4}$$

and both forms are worth having in view: the first is what one applies to a general A, the second is what one reads when a point of $\mathbb{A}^3$ has to be produced or checked by hand. (On the traceless part $a_{22} = -a_{11}$, so $d_a = a_{11} - a_{22} = 2a_{11}$: the quantity d_a of Step 1 is literally twice that entry.) The remaining two coordinates are A's two off-diagonal entries replaced by their sum and difference. No division is needed to **compute** enc – only its inverse (recovering a_{12}, a_{21} individually as $\frac{v_2+v_3}{2}$ and $\frac{v_2-v_3}{2}$) needs 2 invertible.

Step 3: form the commutator. Given traceless $A_0 = \left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$ and $B_0 = \left(\begin{smallmatrix} u & v \\ w & -u \end{smallmatrix}\right)$, the ordinary matrix commutator $[A_0, B_0] = A_0B_0 - B_0A_0$ is again traceless automatically, since $\mathrm{tr}(PQ) = \mathrm{tr}(QP)$ for any P, Q and therefore $\mathrm{tr}([P, Q]) = 0$ always, so it is again encodable by enc.

Step 4: the encoded bracket is the cross product, up to one fixed sign twist.

For the following theorem, the article's standing hypothesis char $\mathbb{F} \neq 2$ is deliberately not used: the identity holds over every commutative ring.

Theorem 1 (the basic identity)

For traceless 2×2 matrices A_0, B_0 over $\mathbb{F}$, with no hypothesis on the characteristic,

$$\mathrm{enc}([A_0, B_0]) = S(\mathrm{enc}(A_0) \times \mathrm{enc}(B_0)), \quad S := \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{5}$$

Proof. The proof is a direct computation, carried out symbolically over generic entries $a_{11}, \ldots, b_{22}$ rather than checked at sampled points, in *code/verify_lorentzian_encoding_identity.sage*. Both sides, expanded in the 8 free entries, agree term by term; they are displayed in full in Section 11.1.

Both sides are integer polynomials in those entries, so the identity descends to every commutative ring. It was also checked directly over $\mathbb{Z}$ and, separately, over $\mathbb{F}_2$ (where it still holds), in *code/referee_round4_setup_determinantal_check.sage*. □

Definition 2 (reduced iterated commutator, signed encoding)

$$\overline{w}_2 := \overline{v}_1 \times \overline{v}_2, \quad \overline{w}_j := S\big(\overline{w}_{j-1}\big) \times \overline{v}_j \quad (j = 3, \ldots, k). \tag{6}$$

Let $R_k := \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$ ($\overline{v}_i \in \mathbb{F}^3$, $3k$ variables) and $I_k \subset R_k$ the ideal generated by the three components of $\overline{w}_k$ – the ideal cutting out $[A_1, \ldots, A_k] = 0$ in reduced coordinates. Two deliberate understatements are worth flagging. First, I_k is at this point only the **generated** ideal; that it is also the full vanishing ideal of that locus is Theorem 39 below, and is not to be assumed before then. Second, the encoded bracket is strictly $S(\overline{w}_k)$, not $\overline{w}_k$ – but S merely negates two of three coordinates, so the two triples generate the same ideal, and it is cleaner to carry the untwisted one.

♣

To spell out exactly what this means: "$R_k := \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$" is shorthand for **the polynomial ring in the $3k$ scalar variables $v_{i,j}$ ($i = 1, \ldots, k$, $j = 1, 2, 3$)**, grouped into k blocks of 3 – **not** a ring with k vector-valued generators (a polynomial ring's generators are always scalars). At $k = 2$, written out in full:

$$R_2 = \mathbb{F}\big[v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}\big], \tag{7}$$

six ordinary scalar indeterminates, with $\overline{v}_1 = \big(v_{1,1}, v_{1,2}, v_{1,3}\big)$ and $\overline{v}_2 = \big(v_{2,1}, v_{2,2}, v_{2,3}\big)$ merely a notational grouping of them into two blocks of three – convenient because $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is most naturally written as a cross product of two block-vectors, not as some formula in six unrelated scalars.

Each $\overline{w}_j$ is a triple of polynomials, i.e. $\overline{w}_j \in R_j^3$ (the free rank-3 module over R_j) – equivalently, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ **given by substitution**, evaluating each of $\overline{w}_j$'s three polynomial components at a point of $\mathbb{A}^{3j}$; and I_k (Definition 2) is, equivalently, the ideal generated by the three coordinate functions of $\overline{w}_k$.

Example 3 ($\overline{w}_k$ and I_k, explicitly, for $k = 2, 3, 4$)

Everything below was computed directly (*code/compute_signed_encoding_examples.sage*) rather than derived by hand. At $k = 2$, the three components of $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$:

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{8}$$

At $k = 3$, the three components of $\overline{w}_3 = S(\overline{w}_2) \times \overline{v}_3$ – with **every coefficient equal to ± 1**, no factor of 2 anywhere:

$$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3}, \tag{9}$$

$$-v_{1,2}v_{2,1}v_{3,1} + v_{1,1}v_{2,2}v_{3,1} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3}, \tag{10}$$

$$v_{1,3}v_{2,1}v_{3,1} - v_{1,1}v_{2,3}v_{3,1} + v_{1,3}v_{2,2}v_{3,2} - v_{1,2}v_{2,3}v_{3,2}, \tag{11}$$

and $I_3 := \big(w_{3,1}, w_{3,2}, w_{3,3}\big)$. At $k = 4$, $\overline{w}_4 = S(\overline{w}_3) \times \overline{v}_4$: again every coefficient is ± 1; the first component of $\overline{w}_4$ is

$$-v_{1,3}v_{2,1}v_{3,1}v_{4,2} + v_{1,1}v_{2,3}v_{3,1}v_{4,2} - v_{1,3}v_{2,2}v_{3,2}v_{4,2} + v_{1,2}v_{2,3}v_{3,2}v_{4,2} \tag{12}$$

$$+v_{1,2}v_{2,1}v_{3,1}v_{4,3} - v_{1,1}v_{2,2}v_{3,1}v_{4,3} + v_{1,3}v_{2,2}v_{3,3}v_{4,3} - v_{1,2}v_{2,3}v_{3,3}v_{4,3}, \tag{13}$$

the other two of the same shape (omitted for space; all three components of $\overline{w}_4$, and $I_4 := \big(w_{4,1}, w_{4,2}, w_{4,3}\big)$, in *code/compute_signed_encoding_examples.sage*'s output). The same data at $k = 6$, where the generators reach 32 terms apiece and the ± 1 pattern is still unbroken, is in Section 11.3.

◇

The displayed examples above were computed over $\mathbb{Q}$; every result below holds over the arbitrary field $\mathbb{F}$ of characteristic $\neq 2$ fixed at the start of this section.

3. The determinantal structure and the generator count

Theorem 4 (determinantal structure)

For $k \geq 3$, I_k is the ideal of 2×2 minors of the 2×3 matrix

$$\begin{pmatrix} S(\overline{w}_{k-1}) \\ \overline{v}_k \end{pmatrix}, \tag{14}$$

i.e. $V(I_k)$ is the preimage, under

$$(\overline{v}_1, \ldots, \overline{v}_k) \mapsto \begin{pmatrix} S(\overline{w}_{k-1}(\overline{v}_1, \ldots, \overline{v}_{k-1})) \\ \overline{v}_k \end{pmatrix}, \tag{15}$$

of the classical rank-≤ 1 determinantal variety of 2×3 matrices. (At $k = 2$ the same statement holds with $S(\overline{w}_{k-1})$ replaced by $\overline{v}_1$ itself, matching Definition 2′s base case $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ directly.)

Proof. The claim is immediate from two facts taken together. The three components of a cross product $a \times b$ are, up to sign, the 2×2 minors of the 2×3 matrix with rows a and b; and Definition 2′s recursion builds $\overline{w}_k$ from exactly such a cross product. The statement was also checked independently at $k = 2, \ldots, 6$ in *code/verify_determinantal_structure_all_k.sage*, in two separate ways. First, I_k and the ideal of the displayed minors are shown equal by reducing each ideal's generators to zero against the other's Gröbner basis, in **both** directions, rather than by containment one way. Second – and this is what makes "independent" mean something here – the recursion itself is checked against honest 2×2 matrix multiplication: the A_i are reconstructed from their enc-coordinates, the iterated commutator is formed by repeated matrix products, and the result encoded and compared against $S(\overline{w}_k)$. The two routes share no code. The underlying polynomial identity $\mathrm{enc}([A, B]) = S(\mathrm{enc}(A) \times \mathrm{enc}(B))$ of Theorem 1 makes this a two-line induction valid for **every** k, not merely a pattern checked at small k (*code/verify_lorentzian_encoding_identity.sage*). □

This theorem is the engine behind the geometric results of Section 4: R_k/I_k is not merely **determinantal-looking**, it is literally an iterated extension of the coordinate ring of a classical determinantal variety.

Example 5 (the 2×3 matrix of Theorem 4, for $k = 2, 3, 4$)

At $k = 2$:

$$\begin{pmatrix} \overline{v}_1 \\ \overline{v}_2 \end{pmatrix} = \begin{pmatrix} v_{1,1} & v_{1,2} & v_{1,3} \\ v_{2,1} & v_{2,2} & v_{2,3} \end{pmatrix}. \tag{16}$$

At $k = 3$, the first row is $S(\overline{w}_2) = (-w_{2,1}, -w_{2,2}, w_{2,3})$ where $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is as in the previous example, giving

$$\begin{pmatrix} S(\overline{w}_2) \\ \overline{v}_3 \end{pmatrix} = \begin{pmatrix} v_{1,3}v_{2,2} - v_{1,2}v_{2,3} & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} & -v_{1,2}v_{2,1} + v_{1,1}v_{2,2} \\ v_{3,1} & v_{3,2} & v_{3,3} \end{pmatrix}. \tag{17}$$

At $k = 4$ the first row is $S(\overline{w}_3)$, whose three entries are cubic with four terms each. Naming them A, B, C keeps the matrix legible:

$$\begin{pmatrix} A & B & C \\ v_{4,1} & v_{4,2} & v_{4,3} \end{pmatrix}, \tag{18}$$

where

$$\begin{aligned} A &= -v_{1,2}v_{2,1}v_{3,2} + v_{1,1}v_{2,2}v_{3,2} + v_{1,3}v_{2,1}v_{3,3} - v_{1,1}v_{2,3}v_{3,3}, \\ B &= v_{1,2}v_{2,1}v_{3,1} - v_{1,1}v_{2,2}v_{3,1} + v_{1,3}v_{2,2}v_{3,3} - v_{1,2}v_{2,3}v_{3,3}, \\ C &= v_{1,3}v_{2,1}v_{3,1} - v_{1,1}v_{2,3}v_{3,1} + v_{1,3}v_{2,2}v_{3,2} - v_{1,2}v_{2,3}v_{3,2}. \end{aligned} \tag{19}$$

The row degree grows by one at each step $(1, 2, 3, \ldots)$, while the second row is always the fresh, free vector $\overline{v}_k$ – exactly the "one fixed complicated row, one free row" shape Proposition 26 needs.

Lemma 6 (non-degeneracy, with an explicit witness)

For every $j \geq 2$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ is not identically zero. Concretely, the point

$$p^{(j)} := \left(e_1, e_2, \underbrace{e_1, \ldots, e_1}_{j-2} \right) \in \mathbb{A}^{3j} \tag{20}$$

satisfies $\overline{w}_j\big(p^{(j)}\big) \neq 0$; explicitly $\overline{w}_j\big(p^{(j)}\big) = e_3$ for j even and e_2 for j odd. Hence $U_j := \big\{p \in \mathbb{A}^{3j} : \overline{w}_j(p) \neq 0\big\}$ is a nonempty Zariski-open subset of $\mathbb{A}^{3j}$.

Proof. The induction is on j, and it carries more than the statement "some point survives": it carries a **named** point, because Lemma 37 later needs one rather than a mere existence claim.

Base case $j = 2$. By Definition 2, $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$, so $\overline{w}_2(e_1, e_2) = e_1 \times e_2 = e_3 \neq 0$. Thus $p^{(2)} = (e_1, e_2)$ works.

Induction step. Let $j \geq 3$ and suppose $p \in \mathbb{A}^{3(j-1)}$ satisfies $c := \overline{w}_{j-1}(p) \neq 0$. The step rests on three observations, taken in order:

1. $S(c) \neq 0$, since $\det S = 1$ makes S invertible over every field, so it kills no nonzero vector.
2. Therefore $\mathbb{F} \cdot S(c)$ is a **line** in $\mathbb{F}^3$. The three vectors e_1, e_2, e_3 span $\mathbb{F}^3$, and a line contains at most one of them; so at least two indices i have $e_i \notin \mathbb{F} \cdot S(c)$. Fix any such i.
3. A cross product vanishes exactly on linearly dependent pairs ($\ker [u]_\times = \mathbb{F}u$ for $u \neq 0$, the computation carried out in the proof of Lemma 38), so $S(c) \times e_i \neq 0$.

By Definition 2, $\overline{w}_j(p, e_i) = S\big(\overline{w}_{j-1}(p)\big) \times e_i = S(c) \times e_i \neq 0$. (The twist falls on c rather than on the result because that is where Definition 2's recursion puts it.) So $\overline{w}_j$ is not identically zero, and appending e_i to a witness at level $j-1$ gives one at level j.

The witness in closed form. Running the step from $p^{(2)} = (e_1, e_2)$ and taking the least admissible i each time, the choice is $i = 1$ at **every** level: $\overline{w}_2 = e_3$ and $S(e_3) = e_3$, which is not proportional to e_1, giving $\overline{w}_3 = e_3 \times e_1 = e_2$; then $S(e_2) = -e_2$, again not proportional to e_1, giving $\overline{w}_4 = (-e_2) \times e_1 = e_3$; and the pair (e_3, e_2) now repeats. Hence the closed form in the statement, and every coordinate of both the witness and its image lies in $\{0, \pm 1\}$. This has been verified for $j = 2, \ldots, 8$, by two independent evaluations per level, in *code/verify_nondegeneracy_witness.sage*, and the results are tabulated in Section 11.4.

Openness. U_j is the complement of the Zariski-closed set $V\big(w_{j,1}, w_{j,2}, w_{j,3}\big)$, hence open; it is nonempty because $p^{(j)}$ lies in it. □

The generator count of I_k needs nothing beyond the definition and non-degeneracy – not codimension, not the determinantal-structure theorem above, not any external citation:

Lemma 7 (independent same-degree generators are minimal)

If a homogeneous ideal in a graded ring is generated by finitely many forms of the same degree that are linearly independent over the base field, that generating set is minimal.

Proof. A polynomial-coefficient combination reproducing one of the generators from the others, while staying in the same degree, can only use degree-0 (constant) coefficients – so a redundant generator would violate linear independence directly. □

Lemma 8 (the three generators of I_k are independent)

The three components of $\overline{w}_k$ are linearly independent over $\mathbb{F}$, for every $k \geq 2$.

Proof. By induction on k, and the induction hypothesis must be **independence** itself; see Remark 9 for why the weaker non-degeneracy of Lemma 6 does not suffice.

Base case $k = 2$. The three components of $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ are the 2×2 minors of the generic 2×3 matrix with rows $\overline{v}_1, \overline{v}_2$, displayed in Example 3. Each contains a monomial the other two do not: $v_{1,2}v_{2,3}$ occurs only in the first, $v_{1,3}v_{2,1}$ only in the second, and $v_{1,1}v_{2,2}$ only in the third. So no nontrivial constant relation is possible.

Induction step. Suppose the components of $\overline{w}_{k-1}$ are linearly independent over $\mathbb{F}$, and let $c \in \mathbb{F}^3$ satisfy $c \cdot \overline{w}_k = 0$. Writing $a := S(\overline{w}_{k-1})$, the scalar triple product identity gives

$$0 = c \cdot (a \times \overline{v}_k) = (c \times a) \cdot \overline{v}_k. \tag{21}$$

The variables $\overline{v}_k$ are a **fresh** block, occurring in neither a nor c, and the right-hand side is linear in them; so each of its three coefficients vanishes separately, i.e.

$$c \times a = 0. \tag{22}$$

Suppose $c \neq 0$. Then $c \times a = 0$ forces $a = f \cdot c$ for a single polynomial f: over the fraction field $c \times a = 0$ says a lies in the line spanned by c, and $f = a_i/c_i$ is a polynomial for any index with $c_i \neq 0$. But then a's three components are pairwise proportional, hence linearly **dependent** over $\mathbb{F}$ (any d with $d \cdot c = 0$ gives $d \cdot a = (d \cdot c)f = 0$, and such a nonzero d exists since c spans only a line in $\mathbb{F}^3$). Since S is invertible, the components of $\overline{w}_{k-1} = S^{-1}(a)$ are then linearly dependent too, contradicting the induction hypothesis. Hence $c = 0$.

Corroborated computationally: the coefficient matrix has rank 3 at $k = 2, \ldots, 6$ (*code/verify_generator_independence.sage*); the full table is in Section 11.5.

This lemma is also **formally verified**: it is the statement *w_linearIndependent* of the Lean 4 / mathlib development accompanying this article, proved there for every $k \geq 2$ over an arbitrary field rather than checked at small k. Theorem 10 below is formalized too, as *w_minimal_generators*. Section 9 says what the formalization does and does not cover. □

Remark 9 (why non-degeneracy is not enough here)

It is tempting to argue instead that a constant relation would force $S(\overline{w}_{k-1})$ to be a **constant vector**, and to rule that out by Lemma 6. That argument is not valid. The relation forces only $c \times S(\overline{w}_{k-1}) = 0$, which makes the components of $S(\overline{w}_{k-1})$ pairwise proportional, not constant. A vector such as $(f, 2f, 3f)$ with $f \neq 0$ is nonvanishing, so Lemma 6 does not exclude it, yet its components are visibly dependent. Non-degeneracy is strictly weaker than what the step needs, which is why the induction above carries linear independence itself.

Theorem 10 (I_k has exactly 3 minimal generators)

For every $k \geq 2$: I_k has exactly 3 minimal generators.

Proof. Immediate from Lemma 7 applied to Lemma 8's independence, since I_k is generated by the three (same-degree-k-homogeneous) components of $\overline{w}_k$ by definition. □

4. The geometric structure of $V(I_k)$

Lemma 11 (continuous images and closures preserve irreducibility)

Let $f : X \to Y$ be a continuous map of topological spaces with X irreducible. Then $f(X)$ (with the subspace topology) is irreducible, and hence so is $\overline{f(X)}$.

Proof. **Image.** Suppose $f(X) = C_1 \cup C_2$ with C_1, C_2 closed in $f(X)$. By continuity, $f^{-1}(C_1)$ and $f^{-1}(C_2)$ are closed in X, and $X = f^{-1}(C_1) \cup f^{-1}(C_2)$. Since X is irreducible, one of the two equals X – say $f^{-1}(C_1) = X$ – so $f(X) \subseteq C_1$, i.e. $C_1 = f(X)$, which is not proper. Hence $f(X)$ is irreducible. **Closure.** If $A \subseteq Y$ is irreducible and $\overline{A} = C_1 \cup C_2$ with C_1, C_2 closed in Y, then $A = (A \cap C_1) \cup (A \cap C_2)$; irreducibility of A forces, say, $A \subseteq C_1$; since C_1 is closed, $\overline{A} \subseteq C_1$, and combined with $C_1 \subseteq \overline{A}$, $C_1 = \overline{A}$, not proper. Apply both parts with $A := f(X)$. □

This is purely topological – continuity of f is the only hypothesis used, nothing about f being a polynomial map or morphism of varieties [11].

Definition 12 (the strata Y_1, Y_2)

For $k \geq 3$, write π for the projection **forgetting $\overline{v}_k$**,

$$\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}, \quad \pi(\overline{v}_1, \ldots, \overline{v}_k) := (\overline{v}_1, \ldots, \overline{v}_{k-1}), \tag{23}$$

and Φ for the map

$$\Phi : \mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}, \quad \Phi(p, \lambda) := (p, \lambda S(\overline{w}_{k-1}(p))). \tag{24}$$

Define two subsets of $\mathbb{A}^{3k}$:

$$Y_1 := V(I_{k-1}) \times \mathbb{A}^3, \quad Y_2 := \overline{\Phi(\mathbb{A}^{3k-2})}. \tag{25}$$

Here Y_1 is the locus where the previous bracket already vanishes, with $\overline{v}_k$ completely free, while Y_2 is the closure of the locus swept out as $\overline{v}_k$ ranges over scalar multiples of $S(\overline{w}_{k-1}(p))$, over all $p \in \mathbb{A}^{3(k-1)}$. At $k = 2$ there is no "previous bracket" to chain from, so neither Y_1 nor Φ is defined; by convention $Y_2 := V(I_2)$ directly there, matching Theorem 4's own base-case treatment.

Lemma 13 (Y_2 is irreducible)

For $k \geq 3$: Y_2 (Definition 12) is irreducible, and in fact its defining ideal (the graph ideal of Φ with λ eliminated) is prime.

Proof. The map Φ is polynomial, hence continuous for the Zariski topology, since preimages of Zariski-closed sets under a polynomial map are again Zariski-closed by substitution. Its domain $\mathbb{A}^{3k-2}$ is irreducible, because the polynomial ring over $\mathbb{F}$ is a domain and so its zero ideal is prime. Lemma 11 then gives Y_2 irreducible directly. Primality of the elimination ideal follows the same way: it is the kernel of a map into a domain ($\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]/(\cdot) \cong \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_{k-1}, \lambda]$), hence prime – this holds for every k by this argument alone, additionally corroborated computationally at $k = 4$ by computing the elimination ideal directly and confirming primality in Sage, as an independent check rather than a substitute for the argument. □

Example 14 (the defining ideals of Y_1 and Y_2, for $k = 2, 3, 4$)

Writing $\overline{v}_i = (v_{i,1}, v_{i,2}, v_{i,3})$ for the three coordinates of block $\overline{v}_i$: at $k = 2$, $Y_2 := V(I_2)$ (the base-case convention above), so its ideal **is** I_2, generated by

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{26}$$

(Y_1 is not defined at $k = 2$, Definition 12.) For $k \geq 3$, Y_1's defining ideal is simply I_{k-1} itself, viewed inside R_k by leaving $\overline{v}_k$ free – at $k = 3$, exactly the three generators displayed above for I_2, now inside R_3 with $\overline{v}_3$ not appearing at all; at $k = 4$, exactly I_3's three generators (displayed below in this same example) inside R_4 with $\overline{v}_4$ free. Unlike Y_2, no elimination is needed to compute Y_1's ideal – it is already I_{k-1}'s own generating set, unchanged. At $k = 3$, Y_2's elimination ideal (computed directly, not inferred) is generated by

$$v_{2,1}v_{3,1} + v_{2,2}v_{3,2} - v_{2,3}v_{3,3}, \quad v_{1,1}v_{3,1} + v_{1,2}v_{3,2} - v_{1,3}v_{3,3}, \tag{27}$$

$$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3} \tag{28}$$

– and these three repay a careful reading, because their shape is not the one the parametrization first suggests. **All three** are linear in the new block $\overline{v}_3$; what separates them is total degree, $2, 2, 3$. The two quadratic ones are, up to sign, the incidence conditions

$$\overline{v}_1 \cdot S(\overline{v}_3) = 0, \quad \overline{v}_2 \cdot S(\overline{v}_3) = 0. \tag{29}$$

These are **consequences** of the proportionality $\overline{v}_3 \propto S(\overline{w}_2)$, not its components: substituting $\overline{v}_3 = \lambda S(\overline{w}_2)$ and using $S^2 = I$ turns them into $\lambda(\overline{v}_i \cdot \overline{w}_2) = 0$, which holds because $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is orthogonal to both of its factors. The proportionality condition itself – the vanishing of $\overline{v}_3 \times S(\overline{w}_2)$, whose three components are, up to sign, exactly the components of $\overline{w}_3$ – enters through a single representative: the third generator, of degree 3, is verbatim the first component of $\overline{w}_3$, and the other two components of $\overline{w}_3$ also lie in this ideal (as they must, since $I_3 \subseteq P_3$; confirmed by direct reduction) without being needed as generators. Note that no generator of this ideal could avoid $\overline{v}_3$ altogether: Y_2 projects dominantly onto $\mathbb{A}^6$, so the elimination ideal meets R_2 in 0. **This ideal strictly contains I_3** – I_3 is its intersection with a second prime (the "Y_1" piece, where the $k = 2$ bracket already vanishes); see Example 25, where this exact ideal reappears as one of I_3's two primary components. At $k = 4$ the same pattern repeats one degree higher, the three generators now having degrees $2, 3, 4$ and again all three linear in the new block $\overline{v}_4$. The first two are once more incidence conditions forced by $\overline{w}_3 \perp S(\overline{w}_2), \overline{v}_3$, namely $\overline{v}_3 \cdot S(\overline{v}_4) = 0$ and $\overline{w}_2 \cdot \overline{v}_4 = 0$; the third is the negative of $\overline{w}_4$'s first component:

$$v_{3,1}v_{4,1} + v_{3,2}v_{4,2} - v_{3,3}v_{4,3}, \tag{30}$$

$$v_{1,3}v_{2,2}v_{4,1} - v_{1,2}v_{2,3}v_{4,1} - v_{1,3}v_{2,1}v_{4,2} + v_{1,1}v_{2,3}v_{4,2} + v_{1,2}v_{2,1}v_{4,3} - v_{1,1}v_{2,2}v_{4,3}, \tag{31}$$

$$v_{1,3}v_{2,1}v_{3,1}v_{4,2} - v_{1,1}v_{2,3}v_{3,1}v_{4,2} + v_{1,3}v_{2,2}v_{3,2}v_{4,2} - v_{1,2}v_{2,3}v_{3,2}v_{4,2} \tag{32}$$

$$-v_{1,2}v_{2,1}v_{3,1}v_{4,3} + v_{1,1}v_{2,2}v_{3,1}v_{4,3} - v_{1,3}v_{2,2}v_{3,3}v_{4,3} + v_{1,2}v_{2,3}v_{3,3}v_{4,3}. \tag{33}$$

All computed directly (elimination of λ from the graph ideal of Φ, in Sage), not just asserted from the pattern. ◇

Lemma 15 (dimension of Y_2, and where it lives)

With Y_2 as in Lemma 13 and U_{k-1} as in Lemma 6: $\dim Y_2 = 3k - 2$, and Y_2 meets $\pi^{-1}(U_{k-1})$ nonemptily, where π is the projection of Definition 12, forgetting $\overline{v}_k$. ♡

Proof. The projection $\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ discards the last block, so that $\pi(\overline{v}_1, \ldots, \overline{v}_k) = (\overline{v}_1, \ldots, \overline{v}_{k-1})$. Hence $\pi^{-1}(U_{k-1}) = U_{k-1} \times \mathbb{A}^3$, the set of points whose **first** $k - 1$ blocks lie in U_{k-1} and whose $\overline{v}_k$ is completely unconstrained. Over U_{k-1}, Φ is injective. Both p and λ can be recovered from $\Phi(p, \lambda)$: p directly, and λ by dividing any nonzero coordinate of $\overline{v}_k = \lambda S(\overline{w}_{k-1}(p))$ by the corresponding coordinate of $S(\overline{w}_{k-1}(p))$; this division is well-defined since $S(\overline{w}_{k-1}(p)) \neq 0$ on U_{k-1}. Injectivity gives $\dim Y_2 = \dim(U_{k-1} \times \mathbb{A}^1) = 3(k-1) + 1 = 3k - 2$. Since U_{k-1} is nonempty by Lemma 6, $\Phi(U_{k-1} \times \{1\})$ is a nonempty subset of $Y_2 \cap \pi^{-1}(U_{k-1})$. □

Theorem 16 (I_k has codimension 2)

For every $k \geq 2$: $\dim V(I_k) = 3k - 2$, i.e. $\operatorname{codim}(I_k) = 2$. ♡

Proof. The proof is by induction on k. **Base case $k = 2$**: $V(I_2)$ is the rank-≤ 1 locus of a 2×3 matrix with independent-indeterminate entries. Its dimension is the standard $4 = 3 \cdot 2 - 2$ for a generic determinantal variety of this shape: the classical codimension formula for the rank-$\leq t$ locus of a generic $m \times n$ matrix gives $(m-t)(n-t) = 1 \cdot 2 = 2$ here [1], and the same value falls out of viewing $V(I_2)$ as the affine cone over the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$, of dimension $1 + 2 + 1 = 4$. The value is confirmed directly by Gröbner-basis computation as well (Example 17). **Inductive step**: by Theorem 4, and since $a \times b = 0$ if and only if a, b are linearly dependent, $V(I_k) = Y_1 \cup Y_2$ (Definition 12). The locus $S(\overline{w}_{k-1}) = 0$ gives Y_1, with $\overline{v}_k$ unconstrained; the locus $S(\overline{w}_{k-1}) \neq 0$ with $\overline{v}_k$ proportional to it gives points **literally in Φ's image**, no closure needed, hence in Y_2. For the reverse containment $Y_1 \cup Y_2 \subseteq V(I_k)$: $\overline{w}_k = S(\overline{w}_{k-1}) \times \overline{v}_k$ vanishes wherever $\overline{w}_{k-1}$ does, so $Y_1 \subseteq V(I_k)$; and $\Phi(\cdot) \subseteq V(I_k)$ identically, since a vector crossed with its own scalar multiple vanishes, so $Y_2 = \overline{\Phi(\cdot)} \subseteq V(I_k)$, the vanishing locus being closed. By the inductive hypothesis, $\dim V(I_{k-1}) = 3(k-1) - 2$, so $\dim Y_1 = 3(k-1) - 2 + 3 = 3k - 2$. Also $\dim Y_2 = 3k - 2$, by Lemma 15. Hence $\dim V(I_k) = \dim(Y_1 \cup Y_2) = 3k - 2$. This dimension statement alone needs no containment argument; that is needed only for the sharper component-count result below. □

Example 17 (dimension and codimension, for $k = 2, 3, 4$)

Values of $\dim R_k$ (the ambient dimension, $= 3k$), $\dim(R_k/I_k)$ (the dimension of the affine variety $V(I_k)$ – **not** "$\dim I_k$", an ideal is not a variety and does not itself have a dimension in this sense), and $\operatorname{codim}(I_k) := \dim R_k - \dim(R_k/I_k)$:

k	$\dim R_k$	$\dim(R_k/I_k)$	$\operatorname{codim}(I_k)$
2	6	4	2
3	9	7	2
4	12	10	2

All three rows follow directly from Theorem 16's formula $\dim(R_k/I_k) = 3k - 2$; also confirmed by direct Gröbner-basis dimension computation in Sage at each k, as an independent check. ◇

Proposition 18 ($Y_2 \not\subseteq Y_1$)

With Y_1, Y_2 as in the proof of Theorem 16: $Y_2 \not\subseteq Y_1$. ♠

Proof. Pick $p \in U_{k-1}$ (Lemma 6); then $\Phi(p, 0) = (p, 0) \in Y_2$ but $\overline{w}_{k-1}(p) \neq 0$ puts $(p, 0) \notin Y_1$. □

Proposition 19 (no component of Y_1 lies in Y_2)

Let Z be an irreducible component of $V(I_{k-1})$ **of dimension $3(k-1) - 2$**. Then $Z \times \mathbb{A}^3 \not\subseteq Y_2$.

The dimension hypothesis is stated rather than cited. Theorem 16 gives the dimension of the **union** $V(I_{k-1})$, which does not by itself say that every component attains it; the per-component statement is part of Theorem 20, and this proposition is used inside that theorem's own induction, where it is available as the inductive hypothesis at level $k - 1$. ♠

Proof. Both $Z \times \mathbb{A}^3$ and Y_2 are irreducible: Y_2 by Lemma 13, and $Z \times \mathbb{A}^3$ because a product of irreducible varieties is irreducible (Z is irreducible by hypothesis, and $\mathbb{A}^3$ trivially is). Both have the **same** dimension $3k - 2$: $Z \times \mathbb{A}^3$ by the hypothesis on Z, and Y_2 by Lemma 15. Containment would therefore force equality. This is refuted: $Z \times \mathbb{A}^3$ is disjoint from $\pi^{-1}(U_{k-1})$, since $\overline{w}_{k-1}$ vanishes identically on $Z \subset V(I_{k-1})$; but Y_2 meets $\pi^{-1}(U_{k-1})$, by Lemma 15. □

Theorem 20 (component count)

For every $k \geq 2$: $V(I_k)$ has exactly $k - 1$ irreducible components, each of dimension $3k - 2$. The component indexed $j = 2, \ldots, k$ involves exactly the first j coordinate blocks and nothing later. ♡

Proof. The proof is by induction on k. **Base case $k = 2$**: the rank-≤ 1 locus of a generic 2×3 matrix is irreducible – the affine cone over the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$, equivalently the Motzkin-Taussky [3] irreducibility of the 2×2 commuting variety – so exactly 1 component, matching $k - 1 = 1$. **Step**: $V(I_k) = Y_1 \cup Y_2$ as in Theorem 16′s proof. By the inductive hypothesis (this theorem at level $k - 1$, which is where the per-component dimension comes from), $V(I_{k-1})$ has $k - 2$ irreducible components, each of dimension $3(k-1) - 2$, none contained in another; crossing with $\mathbb{A}^3$ makes them Y_1's components, each of dimension $3k - 2$. That is exactly the hypothesis Proposition 19 asks for, so it and Proposition 18 together rule out containment between Y_2 and any of them, in either direction. A finite union of irreducible closed sets with no containments is irredundant, giving exactly $(k - 2) + 1 = k - 1$ components. □

This gives I_k's minimal primes and their count, but not yet that I_k is radical (so not yet that these are literally the components of a **primary** decomposition of I_k itself, as opposed to of $\sqrt{I_k}$) – that is closed in Section 6, after Cohen-Macaulayness is available.

Definition 21 (the components P_j)

For $k \geq 2$ and $j = 2, \ldots, k$, let Φ_j be Definition 12′s map Φ run at parameter j (with the base convention $\Phi_2(p, \lambda) := (p, \lambda p)$), and let

$$\Psi_j : \mathbb{A}^{3(j-1)} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)} \to \mathbb{A}^{3k}, \quad (p, \lambda, q) \mapsto \big(\Phi_j(p, \lambda), q\big) \tag{34}$$

be Φ_j with the remaining blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$ carried along untouched. Since Ψ_j is a polynomial map it **is**, read backwards, a homomorphism of $\mathbb{F}$-algebras

$$\varphi_j : R_k \to \mathbb{F}[p, \lambda, q] = \mathbb{F}[t_1, \ldots, t_{3k-2}], \tag{35}$$

sending each variable $v_{i,c}$ to the corresponding coordinate polynomial of Ψ_j. Define

$$P_j := \ker \varphi_j \subset R_k. \tag{36}$$

Two consequences come for free. First, each P_j is **prime**, being the kernel of a homomorphism into a domain – for every j including $j = 2$, with no case distinction and no appeal to Lemma 13, which is stated only for $k \geq 3$. Second, P_j does not depend on which $k \geq j$ it is viewed inside of: enlarging k appends free variables to source and target alike.

Why the kernel rather than the vanishing ideal. It is tempting to define P_j as the vanishing ideal of the j-th component of $V(I_k)$, and over $\overline{\mathbb{F}}$ the two agree – Proposition 23 identifies that component as the closure of Ψ_j's image. But over a **finite** field "vanishing ideal" is the wrong notion outright: $V(I_k)(\mathbb{F})$ is then a finite set of points, and $x^q - x$ vanishes on all of $\mathbb{A}^1(\mathbb{F}_q)$ without being the zero polynomial. The kernel is the right object over every field at once, is well defined with no convention attached, and is what makes the base-change statement in Remark 40 an identity rather than a coincidence.

♣

Example 22 (a suitable point of Y_2, concretely, at $k = 3$)

It is worth having a single explicit point, and not only the defining ideal of Example 25 below. Take $\overline{v}_1 = (1, 0, 0)$ and $\overline{v}_2 = (0, 1, 0)$, so that $\overline{v}_1 \in U_2$ by Lemma 6, giving $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2 = (0, 0, 1)$ and $S(\overline{w}_2) = (0, 0, 1)$. Taking $\lambda = 1$ in Definition 12′s Φ, i.e. $\overline{v}_3 := S(\overline{w}_2) = (0, 0, 1)$, gives the point

$$(\overline{v}_1, \overline{v}_2, \overline{v}_3) = ((1, 0, 0), (0, 1, 0), (0, 0, 1)) \in Y_2 \subset \mathbb{A}^9. \tag{37}$$

One checks this directly: $\overline{w}_3 = S(\overline{w}_2) \times \overline{v}_3 = (0, 0, 1) \times (0, 0, 1) = 0$, a vector crossed with itself. So the point lies in $V(I_3)$, as any point of the form $\Phi(p, \lambda)$ must (Definition 12). The point lies on the **new** component P_3, not on the inherited P_2 – $\overline{v}_1$ and $\overline{v}_2$ are independent here, so the $k = 2$ bracket does not already vanish. It is **not** a rank-2 witness in Lemma 37′s sense (that witness uses $\lambda = 0$, not $\lambda = 1$) – it illustrates Y_2 itself, not the radicality argument's own witness points.

◇

Proposition 23 (rational parametrization of each component)

For $k \geq 2$ and $j = 2, \ldots, k$, the component of $V(I_k)$ indexed j (Theorem 20, over $\overline{\mathbb{F}}$ per Remark 40) is exactly the closure of the image of Definition 21′ map Ψ_j; equivalently, $P_j = \ker \varphi_j$ is its vanishing ideal. In particular that component is a **rational** variety. The parametrization is moreover **birational** onto the component: Ψ_j is injective over $U_{j-1} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)}$ – where, at $j = 2$, the symbol U_1 is to be read as $\{\overline{v}_1 \neq 0\}$, since Lemma 6 defines U_j only for $j \geq 2$ and $\overline{w}_1$ does not exist.

♠

Proof. At parameter $j = k$ the map Ψ_k is Φ itself, whose image closure is Y_2 (Definition 12), and Theorem 20′s induction identifies Y_2 with the component indexed k. For $j < k$ the same construction run at parameter j gives that level's Y_2, and appending the free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$ crosses it with $\mathbb{A}^{3(k-j)}$ – which is exactly Theorem 20′s description of the inherited component indexed j. That the vanishing ideal of the closure of a polynomial map's image is the kernel of the corresponding algebra homomorphism is the standard identification, over $\overline{\mathbb{F}}$.

For birationality, Lemma 15′s injectivity argument (over U_{j-1}, via Lemma 6 at level $j-1$) applies verbatim at parameter j, and appending the free blocks q unchanged does not affect injectivity. □

Remark 24 (a reducible variety is not itself "rational" – component-by-component is what Proposition 23 answers)

$Y_1 := V(I_{k-1}) \times \mathbb{A}^3$ (Definition 12) is **not** irreducible for $k \geq 4$ – it is a union of $k-2$ of the components above (Theorem 20′s inductive step) – so a single rational parametrization **of Y_1 as one object** is not meaningful in the usual sense (rationality is a property of irreducible varieties). What **is** meaningful, and what Proposition 23 supplies, is a rational parametrization of each of Y_1's irreducible pieces individually – exactly the components inherited from $V(I_{k-1})$.

◇

Example 25 (the components, worked out at $k = 3$)

Uniformly by k, since the components' **count** and **block pattern** are already known (Theorem 20, Definition 21) and only their explicit generators change:

- **At $k = 2$**: I_2 is already prime (Theorem 20′s base case), so $V(I_2)$ has exactly $k - 1 = 1$ component – itself.
- **At $k = 3$**: exactly $k - 1 = 2$ components, both computed directly (Gröbner bases, not inferred from the pattern alone), matching $I_3 = P_2 \cap P_3$:
  - P_2, the "Y_1" piece (block pattern $\{1, 2\}$, matching that the length-2 sub-bracket already vanishes with $\overline{v}_3$ free) – generated by exactly I_2's three generators from the first example, now inside R_3 with $\overline{v}_3$ simply not appearing:
  $$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{38}$$
  - P_3, the "Y_2" piece (block pattern $\{1, 2, 3\}$) – exactly the ideal already computed in the Y_2-defining-ideal example above:
  $$v_{2,1}v_{3,1} + v_{2,2}v_{3,2} - v_{2,3}v_{3,3}, \quad v_{1,1}v_{3,1} + v_{1,2}v_{3,2} - v_{1,3}v_{3,3}, \tag{39}$$
  $$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3}. \tag{40}$$
- **At $k = 4$**: exactly $k - 1 = 3$ components, block patterns $\{1, 2\}$, $\{1, 2, 3\}$, $\{1, 2, 3, 4\}$:
  - The first two, P_2 and P_3, are inherited **verbatim** from $k = 3$ above, with $\overline{v}_4$ simply left free – not recomputed, per Definition 21′ own "does not depend on which k" clause.
  - The third, P_4, is the new "Y_2" stratum – exactly the $k = 4$ row of the earlier Y_2-defining-ideal example (generators omitted here to avoid repeating that display; same growth pattern, quartic with up to 8 terms per generator).

◇

5. Cohen-Macaulayness

Proposition 26 (Eagon-Northcott / Hilbert-Burch, applied)

Let M be a 2×3 matrix with entries in a polynomial ring R over a field (arbitrary entries, not necessarily independent indeterminates), and J the ideal of its 2×2 minors. If J has the **expected** codimension $3 - 2 + 1 = 2$, then J is a perfect ideal of grade 2 – $\mathrm{pd}(R/J) = 2$, hence R/J is Cohen-Macaulay. ♠

Proof. Bruns and Vetter [1], Theorem (2.7), p. 13, gives exactly this for an **arbitrary** matrix, as needed since one row of our 2×3 matrix $(S(\overline{w}_{k-1}))$ is a fixed polynomial vector, not a row of independent indeterminates. (The construction originates with Eagon and Northcott [2]. It is the more general form given by Bruns and Vetter that is applied here.) □

Remark 27 (why the generator count above wasn't proven this way)

One might expect Proposition 26 to hand over the generator count as well, for free, via Bruns-Vetter's Theorem (16.36), p. 217, the Theorem of Hilbert-Burch. It does not. That theorem as actually stated gives $I = a \cdot I_m(f)$ for a nonzerodivisor a, and a generator count is not among its conclusions.

Nor could it be. At expected codimension the minors of an **arbitrary** 2×3 matrix can be perfect and Cohen-Macaulay while one minor vanishes identically, or is a scalar combination of the other two. A single example settles it: over $\mathbb{F}[x, y]$ the matrix $\left(\begin{smallmatrix} x & y & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ has minors $0, x, y$ and height 2, exactly as expected, yet $J = (x, y)$ is perfect and Cohen-Macaulay with only **two** generators.

This is why Theorem 10 is proven directly from linear independence rather than cited from this section's machinery. ♢

Theorem 28 (R_k/I_k is Cohen-Macaulay)

For every $k \geq 2$: R_k/I_k is Cohen-Macaulay. ♡

Proof. By Theorem 16, I_k has codimension 2, the expected value for the 2×3 matrix of Theorem 4 presenting it. Proposition 26 applies directly. (This uses Theorem 16 only – not Theorem 10's generator count, not Theorem 20's component count – genuinely independent of both.) □

6. Radicality

Cohen-Macaulayness and the explicit component structure of Section 4 combine, via Serre's criterion, to give radicality – which makes I_k itself, not just $\sqrt{I_k}$, the object carrying the properties established so far, and upgrades Section 4's component count to a genuine primary decomposition.

Serre's two conditions are written (R_n) and (S_n) throughout, parenthesised to keep them apart from this article's polynomial rings R_k. A Noetherian ring A satisfies (S_n) if $\mathrm{depth}(A_P) \geq \min(n, \dim A_P)$ for every prime $P \subset A$, and (R_n) if A_P is a **regular** local ring for every prime P of height $\leq n$; in particular (R_0) asks for regularity at the minimal primes only. Serre's criterion for reducedness [12] says that a Noetherian ring is reduced if and only if it satisfies both (R_0) and (S_1).

Lemma 29 (reduction via Serre)

R_k/I_k satisfies (S_n) for every n – in particular (S_1). Consequently I_k is radical if and only if R_k/I_k satisfies (R_0): regularity at each of its minimal primes. ♡

Proof. Cohen-Macaulayness (Theorem 28) is the extremal case of (S_n): it asks for $\mathrm{depth}(A_P) = \dim A_P$ at **every** prime, so $\mathrm{depth}(A_P) = \dim A_P \geq \min(n, \dim A_P)$ holds trivially whatever n is, and a Cohen-Macaulay ring satisfies (S_n) for every n. (This needs nothing beyond the definitions, and is standard: Bruns and Vetter [1], or Eisenbud [14].) Given (S_1), Serre's criterion makes reducedness of R_k/I_k – radicality of I_k – equivalent to (R_0) alone. □

The four lemmas that follow are the standard Jacobian criterion, split into its two genuinely separate ingredients – one linear-algebraic (Lemma 30), one dimension-theoretic (Lemma 31) – before either is used. The order is deliberate: the rank bound and the regularity criterion each draw on **both** ingredients and on neither each other.

Lemma 30 (the cotangent identity)

Let $S = \mathbb{F}[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_m)$, $A = S/I$, and $x \in V(I)$ an $\mathbb{F}$-rational point, with $\mathfrak{n} \subset S$ the maximal ideal at x and $\mathfrak{m} = \mathfrak{n}/I \subset A$ its image. Then

$$\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - \operatorname{rank} \operatorname{Jac}(x). \tag{41}$$

Proof. Only linear algebra is needed here; in particular, no hypothesis on A is required. Sending g to its gradient at x identifies the cotangent space $\mathfrak{n}/\mathfrak{n}^2$ with $\mathbb{F}^n$. Under that identification the image of I is the span of the linear parts at x of $f_1, \ldots, f_m$: a general element of I is $\sum_i g_i f_i$, whose linear part at x is $\sum_i g_i(x)$ times that of f_i, since every f_i vanishes at x. That span is the row space of $\operatorname{Jac}(x)$, of dimension $\operatorname{rank} \operatorname{Jac}(x)$. As $\mathfrak{m}/\mathfrak{m}^2 \cong (\mathfrak{n}/\mathfrak{n}^2)/\operatorname{im}(I)$, the claim follows. □

Lemma 31 (the local dimension at a rational point)

Let $S = \mathbb{F}[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_m)$, $A = S/I$ equidimensional of dimension $n - c$, and $x \in V(I)$ an $\mathbb{F}$-rational point with maximal ideal $\mathfrak{m} \subset A$. Write $\operatorname{Jac}(x) \in \mathbb{F}^{m \times n}$ for the Jacobian $\left(\partial f_i/\partial x_j\right)$ **evaluated at** x: a matrix of scalars, not the matrix of polynomials it is evaluated from. (The polynomial Jacobian itself appears later as J_k in Lemma 34; the two are not the same object, and only the evaluated one is used in this section.) Then $\dim A_{\mathfrak{m}} = n - c$.

Proof. The local dimension is computed componentwise:

$$\dim A_{\mathfrak{m}} = \max\{\dim (A/P)_{\mathfrak{m}} : P \in \operatorname{Min}(A), P \subseteq \mathfrak{m}\}, \tag{42}$$

and each $A/P = S/\tilde{P}$ is an affine domain over a field, for which $\dim \left(S/\tilde{P}\right)_{\mathfrak{M}} = \dim S/\tilde{P}$ at **every** maximal ideal $\mathfrak{M}$ – affine domains over a field are catenary and equicodimensional (Eisenbud [14], *Commutative Algebra*, Cor. 13.4; equivalently the Stacks Project, tag 00OS [4] – a dimension formula for finitely generated field extensions, holding over an arbitrary field, not needing a perfect-field hypothesis by accident). By equidimensionality every such $\dim S/\tilde{P}$ equals $n - c$. □

Lemma 32 (Jacobian criterion: the rank bound)

The hypotheses are those of Lemma 31. Then $\operatorname{rank} \operatorname{Jac}(x) \leq c$.

Proof. In any Noetherian local ring one has $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 \geq \dim A_{\mathfrak{m}}$. By Lemma 30 the left-hand side is $n - \operatorname{rank} \operatorname{Jac}(x)$, and by Lemma 31 the right-hand side is $n - c$; rearranging gives $\operatorname{rank} \operatorname{Jac}(x) \leq c$. □

Lemma 33 (Jacobian criterion: regularity at rank $= c$)

The hypotheses are those of Lemma 31. If $\operatorname{rank} \operatorname{Jac}(x) = c$, then $A_{\mathfrak{m}}$ is regular of dimension $n - c$; in particular x lies on a **unique** component of $V(I)$, whose local ring there is a field.

Proof. If $\operatorname{rank} \operatorname{Jac}(x) = c$ then $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - c$ by Lemma 30, while $\dim A_{\mathfrak{m}} = n - c$ by Lemma 31; equality of the two is the definition of regularity. A regular local ring is a domain, so $\mathfrak{m}$ contains a unique minimal prime of A, and the local ring at that prime is a further localization of a regular local ring at its own maximal ideal in dimension 0, hence a field. □

Lemma 34 (the Jacobian's own recursion)

Write J_k for the $3 \times 3k$ Jacobian of $\overline{w}_k$, $[u]_\times$ for the skew matrix with $[u]_\times v = u \times v$. Then $J_2 = \left[-[\overline{v}_2]_\times \;\middle|\; [\overline{v}_1]_\times\right]$, and for $k \geq 3$,

$$J_k = \left[-[\overline{v}_k]_\times S J_{k-1} \;\middle|\; [S(\overline{w}_{k-1})]_\times\right]. \tag{43}$$

Proof. For the second block, $\overline{w}_k = S(\overline{w}_{k-1}) \times \overline{v}_k = [S(\overline{w}_{k-1})]_\times \overline{v}_k$ is linear in $\overline{v}_k$ with coefficient matrix free of $\overline{v}_k$. For the first, differentiating $\overline{w}_k = S(\overline{w}_{k-1}) \times \overline{v}_k$ with respect to any variable t among $\overline{v}_1, \ldots, \overline{v}_{k-1}$ gives $\partial_t \overline{w}_k = (S\partial_t \overline{w}_{k-1}) \times \overline{v}_k = -[\overline{v}_k]_\times S \partial_t \overline{w}_{k-1}$, using $a \times b = -[b]_\times a$; assembling columns gives the stated product. □

Example 35 (J_3, explicitly)

With variables ordered $v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}, v_{3,1}, v_{3,2}, v_{3,3}$, the 3×9 Jacobian of $\overline{w}_3$ (the three generators of I_3 from the earlier example), column-blocked to match Lemma 34's formula $J_3 = \left[-[\overline{v}_3]_\times S J_2 \;\middle|\; [S(\overline{w}_2)]_\times\right]$ – shown here as the two blocks separately, since side by side the full 3×9 matrix does not fit the page width. First, $-[\overline{v}_3]_\times S J_2$ (3×6):

$$\begin{pmatrix} -v_{2,2}v_{3,2} + v_{2,3}v_{3,3} & v_{2,1}v_{3,2} & -v_{2,1}v_{3,3} & v_{1,2}v_{3,2} - v_{1,3}v_{3,3} & -v_{1,1}v_{3,2} & v_{1,1}v_{3,3} \\ v_{2,2}v_{3,1} & -v_{2,1}v_{3,1} + v_{2,3}v_{3,3} & -v_{2,2}v_{3,3} & -v_{1,2}v_{3,1} & v_{1,1}v_{3,1} - v_{1,3}v_{3,3} & v_{1,2}v_{3,3} \\ -v_{2,3}v_{3,1} & -v_{2,3}v_{3,2} & v_{2,1}v_{3,1} + v_{2,2}v_{3,2} & v_{1,3}v_{3,1} & v_{1,3}v_{3,2} & -v_{1,1}v_{3,1} - v_{1,2}v_{3,2} \end{pmatrix} \tag{44}$$

and then $[S(\overline{w}_2)]_\times$ (3×3, the skew matrix of $S(\overline{w}_2)$'s three coordinates from the earlier matrix example):

$$\begin{pmatrix} 0 & v_{1,2}v_{2,1} - v_{1,1}v_{2,2} & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} \\ -v_{1,2}v_{2,1} + v_{1,1}v_{2,2} & 0 & -v_{1,3}v_{2,2} + v_{1,2}v_{2,3} \\ v_{1,3}v_{2,1} - v_{1,1}v_{2,3} & v_{1,3}v_{2,2} - v_{1,2}v_{2,3} & 0 \end{pmatrix} \tag{45}$$

Both blocks computed directly, not merely asserted from the recursive formula. ◇

Remark 36 (a false shortcut)

A naive reading of Lemma 34 suggests appending $\overline{v}_k = 0$ preserves rank on an inherited component – this is false: the old-variable block is left-multiplied by $-[\overline{v}_k]_\times S$, not merely padded with zero rows. At the first inherited case ($k = 2 \to 3$), appending $\overline{v}_3 = 0$ actually drops the rank to 0. The correct witness (Lemma 38 below) instead picks $\overline{v}_k = e_i$ for a standard basis vector **not** lying in $S \cdot \operatorname{im} J_{k-1}(q)$ – a genuine choice, not a formality. ◇

Lemma 37 (rank-2 witness on the new component)

For every $k \geq 3$, there is a point $x = (p, 0)$, p the Lemma 6 non-degeneracy witness at level $k-1$, with $x \in V(I_k)$ and rank $J_{k(x)} = 2$. At $k = 2$ the same conclusion holds with $x = (p, 0)$ for any $p \neq 0$. ♡

Proof. **The case $k = 2$ separately**, because the statement above cannot be read at $k = 2$: it would refer to a non-degeneracy witness "at level 1", and neither $\overline{w}_1$ nor Lemma 6 exists there, since Definition 2 starts the recursion at $j = 2$. Nor does Lemma 34's $k \geq 3$ formula apply. Argue directly instead: $J_2 = \left[-[\overline{v}_2]_\times \;\middle|\; [\overline{v}_1]_\times\right]$ carries no S at all, so for any $p \neq 0$, $J_2(p, 0) = [0 \mid [p]_\times]$, which has rank 2 by the kernel computation below. And $(p, 0) \in V(I_2)$ since $\overline{w}_2(p, 0) = p \times 0 = 0$.

For $k \geq 3$. By Lemma 34, $J_{k(p,0)} = \left[0 \;\middle|\; [S(\overline{w}_{k-1}(p))]_\times\right]$ since $-[0]_\times S J_{k-1}(p) = 0$; and $S(\overline{w}_{k-1}(p)) \neq 0$ since $\overline{w}_{k-1}(p) \neq 0$ (Lemma 6) and S is invertible. For $u \neq 0$, $\ker [u]_\times = \{v : u \times v = 0\} = \mathbb{F}u$ (a cross product vanishes exactly on linearly dependent pairs), so $[S(\overline{w}_{k-1}(p))]_\times$ has rank 2, giving rank $J_{k(p,0)} = 2$. □

Lemma 38 (rank-2 witness on an inherited component)

For every $k \geq 3$ and every component Z of $V(I_{k-1})$ with a rank-2 witness $q \in Z$ (Lemma 37, or this lemma applied at $k-1$), there is $x = (q, e_i) \in Z \times \mathbb{A}^3 \subset V(I_k)$ with rank $J_{k(x)} = 2$. ♡

Proof. Let $W := S \cdot \operatorname{im} J_{k-1}(q)$, a 2-dimensional subspace of $\mathbb{F}^3$ since S is invertible and rank $J_{k-1}(q) = 2$. Since e_1, e_2, e_3 span $\mathbb{F}^3$ and $W \neq \mathbb{F}^3$, some $e_i \notin W$; fix such an i. Since $q \in V(I_{k-1})$, $\overline{w}_{k-1}(q) = 0$, so Lemma 34 gives $J_{k(q,e_i)} = \left[-[e_i]_\times S J_{k-1}(q) \;\middle|\; 0\right]$, of rank $\dim\left([e_i]_\times W\right)$ since $S J_{k-1}(q)$ has image exactly W. As $\ker [e_i]_\times = \mathbb{F}e_i$ and $W \cap \mathbb{F}e_i = 0$ (else $e_i \in W$), $[e_i]_\times$ is injective on W, so $\dim\left([e_i]_\times W\right) = 2$. □

Theorem 39 (I_k is radical)

For every $k \geq 2$: I_k is radical. ♡

Proof. By induction on k (mirroring Theorem 20's own induction), Lemma 37 and Lemma 38 exhibit a rank-2-Jacobian point on **every** one of the $k-1$ components of Theorem 20. Lemma 33 then gives (R_0) at every one (and, by its "unique component" clause, confirms each witness lies only on its intended component – an

independent check on Theorem 20's count, not merely assumed); Lemma 29 gives (S_1), using Theorem 28's Cohen-Macaulayness. Serre's criterion [12] gives radicality. □

This has been verified independently in Sage, for the signed encoding, at every $k = 2, \ldots, 6$: the Jacobian recursion (Lemma 34) checked as a polynomial identity, not just at points, and every witness confirmed to lie on $V(I_k)$ with rank $J_k = 2$ (*code/referee_radicality_S_check.sage*). The sharper claim, that each witness lies on **exactly** its intended component, is confirmed by an independently-computed primary decomposition at every $k = 3, 4, 5, 6$, which returns $k - 1$ components in each case, in agreement with Theorem 20; the $k = 6$ decomposition takes some hours.

7. The Primary Decomposition Theorem

Every ingredient is now in hand for the fully general statement – component count (Section 4) and radicality (Section 6) together, neither sufficient alone. One convention has to be made explicit first, because it is the hinge on which the statement's claimed generality turns.

Remark 40 (the geometric arguments run over $\overline{\mathbb{F}}$, and the conclusion descends)

The three preceding sections argue geometrically: they speak of $V(I_k)$, of its irreducible components, and of the closure of the image of a parametrization, and at the decisive step they identify $\mathrm{Min}(I_k)$ with the vanishing ideals of those components. That identification is a Nullstellensatz statement and is **false over a general field** – over $\mathbb{F}_3$, for instance, $V(I_k)(\mathbb{F}_3)$ is a finite set of points and carries no such information at all. So throughout those sections, and in the theorem below, $V(\cdot)$ means the vanishing locus in $\mathbb{A}^{3k}\left(\overline{\mathbb{F}}\right)$ over a fixed algebraic closure, and "component" means an irreducible component there. This is the usual convention and is what the classical literature on the commuting variety assumes as well (Artin-Hochster is stated over an algebraically closed field in [7]).

The conclusion nonetheless descends to $\mathbb{F}$ itself, so the theorem may be read over any field of characteristic $\neq 2$, as stated. The reason is that each P_j is not merely prime but **absolutely** prime. By Definition 21, P_j **is** the kernel of an $\mathbb{F}$-algebra homomorphism $\varphi_j : R_k \to \mathbb{F}[t_1, \ldots, t_{3k-2}]$ into a polynomial ring, so R_k/P_j embeds in that polynomial ring. Since $\overline{\mathbb{F}}$ is flat over $\mathbb{F}$, tensoring preserves the embedding:

$$\overline{\mathbb{F}} \otimes_{\mathbb{F}} \left(R_k/P_j\right) \hookrightarrow \overline{\mathbb{F}}[t_1, \ldots, t_m], \tag{46}$$

whose target is a domain, so the extension of P_j to $\overline{\mathbb{F}} \otimes_{\mathbb{F}} R_k$ is again prime. Formation of a finite intersection of ideals likewise commutes with the flat base change $\mathbb{F} \to \overline{\mathbb{F}}$. Hence the displayed decomposition, established over $\overline{\mathbb{F}}$, holds already over $\mathbb{F}$, with the same $k - 1$ primes. ◇

Theorem 41 (Primary Decomposition Theorem)

For every $k \geq 2$, over any field of characteristic $\neq 2$: I_k has exactly $k - 1$ irreducible, radical, primary components, namely $P_2, \ldots, P_k$ of Definition 21, and

$$I_k = P_2 \cap \cdots \cap P_k \tag{47}$$

is an irredundant primary decomposition into $k - 1$ primes. ♡

Proof. The two halves were proven independently above, so this combines them in full rather than asserting the conjunction in one line.

The decomposition itself. By Theorem 39, I_k is radical, so $I_k = \sqrt{I_k}$. A radical ideal in a Noetherian ring equals the intersection of its own minimal primes:

$$I_k = \cap_{P \in \mathrm{Min}(I_k)} P. \tag{48}$$

This holds because $\sqrt{I_k}$ is the intersection of **all** primes containing I_k – Krull's theorem, not the definition of the radical, which is instead the set of elements with a power in I_k – and the minimal primes already suffice. Every non-minimal prime containing I_k contains a minimal one, and is therefore redundant in the intersection.

Theorem 20 identifies $\mathrm{Min}(I_k)$ explicitly, over $\overline{\mathbb{F}}$ as Remark 40 requires: exactly $k - 1$ primes, one per block pattern $j = 2, \ldots, k$, named $P_2, \ldots, P_k$ in Definition 21. Substituting them gives the displayed equation.

Why the components are primary. Each P_j is prime, so it is trivially its own primary-decomposition component, being P_j-primary with associated prime P_j. Nothing beyond primality has to be argued.

Why they are prime. Nothing is left to prove here either. Definition 21 **defines** P_j as $\ker \varphi_j$, and the kernel of a homomorphism into a domain is prime. That argument covers $j = 2$ along with the rest, at no extra cost. Lemma 13 would give the same conclusion for $j \geq 3$, but it is stated for $k \geq 3$ and so misses $j = 2$; P_2 is I_2 itself, the ideal of maximal minors of a generic 2×3 matrix, which is classically prime. Neither of those routes is needed.

Extending P_j from R_j to R_k by adjoining the free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$ preserves primality, since R_k/P_jR_k is a polynomial ring over the domain R_j/P_j.

Irredundancy. No P_j can be omitted, i.e. no P_j contains $\cap_{i \neq j} P_i$. This is exactly Theorem 20′s own no-containment conclusion, obtained from Proposition 18 and Proposition 19 applied at every level of that theorem's induction. It needs no further proof here. □

Remark 42 (summary)

Theorem 16 (codimension 2), Theorem 10 (3 minimal generators), Theorem 28 (Cohen-Macaulay), and Theorem 39 (radical) together say exactly what a single bundled "I_k has codimension 2, is radical, has exactly 3 minimal generators, and is Cohen-Macaulay" statement would – now as four independently-proven results, each established at the point in this article where its actual dependencies (and no others) were available, rather than one claim proven in parts across several sections. ◇

8. Summary

For every $k \geq 2$, over any field $\mathbb{F}$ of characteristic $\neq 2$:

- I_k has codimension 2 (Theorem 16), presented as the ideal of 2×2 minors of an explicit 2×3 matrix (Theorem 4) – a genuine determinantal ideal, not merely determinantal-looking.
- I_k has exactly 3 minimal generators (Theorem 10), proven directly from linear independence, independent of the determinantal presentation.
- R_k/I_k is Cohen-Macaulay (Theorem 28), from the determinantal presentation and classical Eagon-Northcott/Hilbert-Burch theory (Proposition 26).
- I_k is radical (Theorem 39), via Serre's criterion: Cohen-Macaulayness gives (S_1) (Lemma 29), and an explicit rank-2 Jacobian witness on every component gives (R_0) (Lemma 37, Lemma 38).
- $V(I_k)$ has exactly $k-1$ irreducible components (Theorem 20), each rational – in fact birational to an explicit affine parameter space (Proposition 23). Radicality and this component count together – rationality plays no part in the assembly – give the irredundant primary decomposition of the Primary Decomposition Theorem (Theorem 41): $I_k = P_2 \cap \cdots \cap P_k$.

9. Methods

This article was produced by a workflow that is not the usual one. Since the workflow bears on how some of the claims above should be read, it is set out here rather than left to be inferred.

Computer algebra, in two roles that should not be confused. SageMath and Macaulay2 were used throughout, for two different purposes. The first is **conjecture**: the component count of Theorem 20, its block-involvement pattern, and the shape of the primary decomposition were found by computing primary decompositions at small k and reading off the pattern, well before any of it was proved. The second is **verification**: once proved, the general statements were checked against independent computation – the determinantal structure by Gröbner-basis equality in both directions and, separately, against ordinary 2×2 matrix multiplication (the two routes sharing no code); the Jacobian recursion as a polynomial identity rather than at sampled points; the witness points for position and rank, on exactly the intended components; and the decomposition itself recomputed at $k = 3, 4, 5, 6$.

Because rank, dimension and radicality are **not** characteristic-independent (a matrix of rank 3 over $\mathbb{Q}$ may drop rank modulo p), computations over $\mathbb{Q}$ license nothing about $\mathbb{F}_p$, and the four main results were therefore re-run over $\mathbb{F}_2$, $\mathbb{F}_3$, $\mathbb{F}_4$, $\mathbb{F}_5$, $\mathbb{F}_7$ and $\mathbb{F}_9$ as well. No deviation appeared. The scripts are named at the point of use, and are deposited, with their full output transcripts, at Zenodo (DOI: 10.5281/zenodo.23433730, with 10.5281/zenodo.23433729 for all versions); the paths cited in the text are the paths inside that archive. The deposit's `runs/REFERENCE-CERTIFICATE.md` records, for each cited source, the exact statement relied on and how it was verified. The explicit computations are displayed in Section 11.

Nothing in this article is conjectural, or checked only at small k: every statement is proven for **every** $k \geq 2$, over an **arbitrary** field of the stated characteristic. The computational thread corroborates the general proofs; nowhere does it substitute for one.

A partial formalization. The statements were formalized in Lean 4 against *mathlib* and machine-checked, with no proof gaps and no axioms beyond the three that every *mathlib* development uses. The formalization reaches part of the article, and the division is this.

Formalized:

- the basic identity (Theorem 1), in the strengthened form that drops the traceless hypothesis;
- the minimal-generator theorem (Theorem 10) for the actual $\overline{w}_k$, with the chain beneath it: that independent same-degree generators are minimal (Lemma 7), the homogeneity of $\overline{w}_k$, the freshness of each new variable block, and the linear independence itself (Lemma 8);
- the Jacobian recursion (Lemma 34);
- assorted smaller computational claims of Section 2 and Section 4.

Not formalized, the libraries for it not presently existing:

- the codimension (Theorem 16);
- Cohen-Macaulayness (Theorem 28);
- radicality (Theorem 39);
- the component count (Theorem 20);
- the Primary Decomposition Theorem (Theorem 41).

Everything in the second list rests on determinantal-ideal theory or on dimension theory. The guarantee is therefore partial, and does not cover the article's main results.

10. Acknowledgements

The author first learned of the ring of commuting matrices some thirty years ago from Professor Jan-Erik Roos, in a course he held on the then fledgling field of computer algebra. It is a pleasant debt to record here, given how much of what follows was found with a computer algebra system open beside the page.

The computations were carried out in SageMath and Macaulay2, and the formalization in Lean 4 with *mathlib*; this article would have a different and smaller content without them. The nature and extent of the assistance from a large language model is set out in Section 9 and in the declaration at the end.

11. Appendix: the explicit computations

The results above are stated with their computations named rather than displayed, so that the argument stays legible. This appendix displays them. Everything here is **corroboration**: no proof in the article rests on a computation, and nothing below is offered as one. Each block names the script that produced it; the scripts and their full transcripts are in the Zenodo deposit (DOI: 10.5281/zenodo.23433730).

(The scripts write `D` internally for the matrix this article calls S – the same matrix, $\mathrm{diag}(-1,-1,1)$.)

11.1. Both sides of the basic identity

Theorem 1 asserts $\mathrm{enc}([A_0,B_0]) = S(\mathrm{enc}(A_0)\times\mathrm{enc}(B_0))$. Expanded over generic entries $a_{11},\ldots,b_{22}$, the two sides are, component by component:

```
lhs_1 = rhs_1 = -2*a21*b12 + 2*a12*b21
lhs_2 = rhs_2 = -a12*b11 + a21*b11 + a11*b12 - a22*b12 - a11*b21 + a22*b21
                + a12*b22 - a21*b22
lhs_3 = rhs_3 = -a12*b11 - a21*b11 + a11*b12 - a22*b12 + a11*b21 - a22*b21
                + a12*b22 + a21*b22
```

The difference is $(0,0,0)$ identically. Chaining once more gives the triple commutator $\mathrm{enc}([[A,B],C])$, which the recursion computes as $S(S(\mathrm{enc}(A)\times\mathrm{enc}(B))\times\mathrm{enc}(C))$; its three components have 8, 20 and 20 terms:

```
comp_1 = -2*a21*b11*c12 + 2*a11*b21*c12 - 2*a22*b21*c12 + 2*a21*b22*c12
         - 2*a12*b11*c21 + 2*a11*b12*c21 - 2*a22*b12*c21 + 2*a12*b22*c21

comp_2 = a12*b11*c11 + a21*b11*c11 - a11*b12*c11 + a22*b12*c11 - a11*b21*c11
         + a22*b21*c11 - a12*b22*c11 - a21*b22*c11 - 2*a21*b12*c12
         + 2*a12*b21*c12 + 2*a21*b12*c21 - 2*a12*b21*c21 - a12*b11*c22
         - a21*b11*c22 + a11*b12*c22 - a22*b12*c22 + a11*b21*c22
         - a22*b21*c22 + a12*b22*c22 + a21*b22*c22

comp_3 = a12*b11*c11 - a21*b11*c11 - a11*b12*c11 + a22*b12*c11 + a11*b21*c11
```

```
        - a22*b21*c11 - a12*b22*c11 + a21*b22*c11 - 2*a21*b12*c12
        + 2*a12*b21*c12 - 2*a21*b12*c21 + 2*a12*b21*c21 - a12*b11*c22
        + a21*b11*c22 + a11*b12*c22 - a22*b12*c22 - a11*b21*c22
        + a22*b21*c22 + a12*b22*c22 - a21*b22*c22
```

The factors of 2 here are expected: on the traceless part $a_{22} = -a_{11}$, so $d_a = a_{11} - a_{22} = 2a_{11}$. The ± 1-coefficient claim of Example 3 concerns the $\overline{v}$-coordinates, a different expansion in different variables. Both routes agree exactly (*code/regenerate_displays_S.sage*, *code/verify_lorentzian_encoding_identity.sage*).

11.2. The sign twist leaves the ideal unchanged

The three components of $\overline{w}_2$, which are also the entries of $S(\overline{w}_2)$ up to sign, are

$$-v_{1,3}v_{2,2} + v_{1,2}v_{2,3}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad -v_{1,2}v_{2,1} + v_{1,1}v_{2,2}, \tag{49}$$

so $\mathrm{ideal}(\overline{w}_2) = \mathrm{ideal}(S\overline{w}_2)$ generator by generator: two of the three are negated and the third is fixed, and negating a generator does not change the ideal it generates – this is Definition 2′s "the two triples generate the same ideal", displayed. Confirmed independently by Gröbner-basis equality in *code/verify_encoding_gl3_equivariance.sage*. Incidentally $S^2 = I$ exactly, so the twist is an involution.

11.3. The generators of I_6

Theorem 10 gives I_k three minimal generators for every k. At $k = 6$ they are degree-6 forms in the 18 variables $v_{1,1}, \ldots, v_{6,3}$, with 32 terms each and **every coefficient equal to ± 1**:

```
w_6,1 = -v1_3*v2_1*v3_1*v4_1*v5_1*v6_2 + v1_1*v2_3*v3_1*v4_1*v5_1*v6_2
        - v1_3*v2_2*v3_2*v4_1*v5_1*v6_2 + v1_2*v2_3*v3_2*v4_1*v5_1*v6_2
        - v1_2*v2_1*v3_2*v4_3*v5_1*v6_2 + v1_1*v2_2*v3_2*v4_3*v5_1*v6_2
        + v1_3*v2_1*v3_3*v4_3*v5_1*v6_2 - v1_1*v2_3*v3_3*v4_3*v5_1*v6_2
        - v1_3*v2_1*v3_1*v4_2*v5_2*v6_2 + v1_1*v2_3*v3_1*v4_2*v5_2*v6_2
        - v1_3*v2_2*v3_2*v4_2*v5_2*v6_2 + v1_2*v2_3*v3_2*v4_2*v5_2*v6_2
        + v1_2*v2_1*v3_1*v4_3*v5_2*v6_2 - v1_1*v2_2*v3_1*v4_3*v5_2*v6_2
        + v1_3*v2_2*v3_3*v4_3*v5_2*v6_2 - v1_2*v2_3*v3_3*v4_3*v5_2*v6_2
        + v1_2*v2_1*v3_1*v4_1*v5_1*v6_3 - v1_1*v2_2*v3_1*v4_1*v5_1*v6_3
        + v1_3*v2_2*v3_3*v4_1*v5_1*v6_3 - v1_2*v2_3*v3_3*v4_1*v5_1*v6_3
        + v1_2*v2_1*v3_2*v4_2*v5_1*v6_3 - v1_1*v2_2*v3_2*v4_2*v5_1*v6_3
        - v1_3*v2_1*v3_3*v4_2*v5_1*v6_3 + v1_1*v2_3*v3_3*v4_2*v5_1*v6_3
        + v1_3*v2_1*v3_1*v4_2*v5_3*v6_3 - v1_1*v2_3*v3_1*v4_2*v5_3*v6_3
        + v1_3*v2_2*v3_2*v4_2*v5_3*v6_3 - v1_2*v2_3*v3_2*v4_2*v5_3*v6_3
        - v1_2*v2_1*v3_1*v4_3*v5_3*v6_3 + v1_1*v2_2*v3_1*v4_3*v5_3*v6_3
        - v1_3*v2_2*v3_3*v4_3*v5_3*v6_3 + v1_2*v2_3*v3_3*v4_3*v5_3*v6_3
```

The other two components, $w_{6,2}$ and $w_{6,3}$, have the same shape and the same 32-term, ± 1 profile; all three are printed in full in *code/regenerate_displays_S.sage*'s transcript, which ships as `runs/display-ground-truth-S.txt`. The script asserts the coefficient profile rather than merely printing it.

11.4. The non-degeneracy witness

The following table evaluates the point constructed in the proof of Lemma 6.

j	witness (v_1,...,v_j)	w_j
2	(e1,e2)	e3
3	(e1,e2,e1)	e2
4	(e1,e2,e1,e1)	e3
5	(e1,e2,e1,e1,e1)	e2
6	(e1,e2,e1,e1,e1,e1)	e3
7	(e1,e2,e1,e1,e1,e1,e1)	e2
8	(e1,e2,e1,e1,e1,e1,e1,e1)	e3

Each level is checked twice in *code/verify_nondegeneracy_witness.sage*: once as the induction runs, and once by feeding the finished witness back through Definition 2′s recursion as a point of $\mathbb{A}^{3j}$, which is how Lemma 6 states it.

11.5. Independence of the generators

This is the corroboration cited in the proof of Lemma 8. For each k, the three components of $\overline{w}_k$ are expanded over the monomials that actually occur and the resulting $3 \times N$ coefficient matrix is checked to have rank 3, which is exactly the assertion that no nontrivial constant relation holds among them:

k	#variables	degree	#monomials	rank	independent?
2	6	2	6	3	yes
3	9	3	12	3	yes
4	12	4	24	3	yes
5	15	5	48	3	yes

6	18	6	96	3	yes

The monomial count doubles at each step while the rank stays at 3, so the margin widens rather than narrows as k grows (*code/verify_generator_independence.sage*).

12. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic), through Claude Code, in order to assist with mathematical exploration, drafting, computational verification and the preparation of the Lean formalization. The models used were Claude Sonnet 5 (August 2026), Claude Opus 5 (August to October 2026) and Claude Opus 5.5 (October 2026), with adversarial reviews of the proofs run with Claude Fable 5 (August and September 2026). After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

The assistance was substantial, and its extent is stated here rather than left to be inferred. The itemised account is Section 9: which results were machine-checked and which were not, what the computer algebra established and what it merely corroborated, and what the adversarial review found. Claude is not a co-author, and is not credited as one.

Every computational claim has been checked along a route independent of the statement being tested, and the reference list has been checked against the sources cited, with the one exception that where a source is a printed book outside this project's open-access policy, an independently verified equivalent is cited alongside it. The author has verified the results to the best of his ability, made the final edits, and assumes full responsibility for any errors, mistakes or gaps that remain.

13. References

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[4] *The Stacks Project*, tag 00OS (dimension formula for finitely generated field extensions / equicodimensionality of affine domains), stacks.math.columbia.edu/tag/00OS.

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[12] *The Stacks Project*, tag **031R** = Lemma 10.157.3, Serre's criterion for reducedness: "Let R be a Noetherian ring. The following are equivalent: (1) R is reduced, and (2) R has properties (R_0) and (S_1)." stacks.math.columbia.edu/tag/031R. Note that the enclosing **section** 10.157 is tag 031O and is titled "Serre's criterion for *normality*"; its own Lemma 10.157.4 (tag 031S) is the normality criterion (R_1) and (S_2), a different statement.

[13] T. Tambour, *The Number of Solutions of Some Equations in Finite Groups and a New Proof of Itô's Theorem*, Communications in Algebra **28** (2000), no. 11, 5353–5361. doi:10.1080/00927870008827160.

[14] D. Eisenbud, *Commutative Algebra with a View Toward Algebraic Geometry*, Graduate Texts in Mathematics 150, Springer-Verlag, 1995. Cor. 13.4 is the statement used here, equivalent to the Stacks Project tag 00OS above.